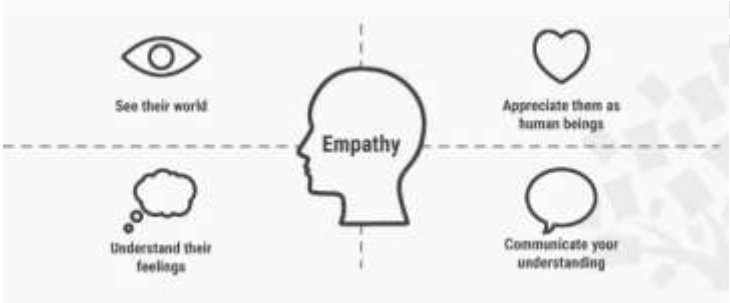
Basics of Interaction Design

Lecture Notes

UNIT-2

Q. 1. What is Empathy in Designing? Why it is so important?

Empathy is our ability to **see the world through other people's eyes** — to see what they see, feel what they feel and experience things as they do. Of course, none of us can fully experience things the way someone else does, but we can attempt to get as close to this as possible. We achieve this empathic state as we put aside our own preconceived ideas about the world and choose to understand the ideas, thoughts and needs of others instead.



In the world of designing, empathy is a "deep understanding of the problems and realities of the people you are designing for." In other words, empathy requires you to learn about the difficulties people face and uncover their latent needs and desires to explain their behaviors. You need to gain an understanding of their environment, as well as their roles and interactions within it if you want to truly empathize with them.

Empathy helps you:

- Appreciate people's emotional and physical needs.
- Gain insight into the way people see, understand and interact with the world around them.
- Realize how lives are impacted within the contexts being investigated.
- Find out what people mean rather than just what they say — empathic research is inherently subjective and is concerned with motivations and thoughts, rather than facts.

Empathy is also the only way to thoroughly understand what people *mean*, rather than just absorb what they *say*. You'll have noticed by now people do *not* necessarily always cover the details when they share stories and other information. They may withhold information out of fear, distrust or another inhibiting factor — be it internal or based on those with whom they engage — and they may express themselves in a less than articulate way. As a designer, you therefore need to:

- Make sense of what is *not* being said, or what is being hinted at beneath the external expressions and words.
- Develop intuition, imagination, emotional sensitivity and creativity so you can dig deeper into people's experiences.
- Extract the right kinds of insight to ensure you make a meaningful difference.

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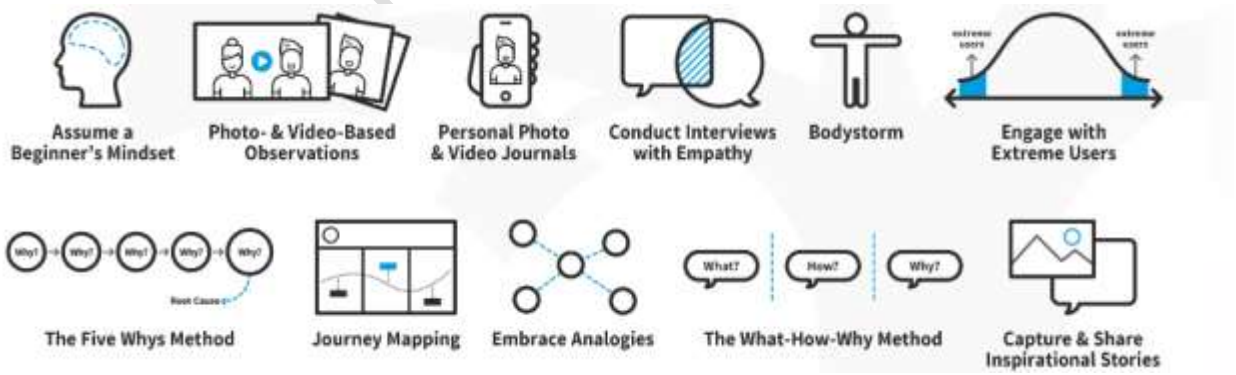
Q. 2 Explain ‘Empathize’ with respect to Design Thinking?

“Empathize” is the first stage of the design thinking process — what could be more important than that?! It makes sense really when you think about it. **Your goal, as a designer, is to gain an empathic understanding of the people you design for, after all!** You should have a burning desire to observe, engage and empathize with the people you design for to understand their experiences and motivations. What’s more, you’ll need to immerse yourself in their physical environment if you want to have any chance at gaining a deeper personal understanding of the issues, needs and challenges involved in their day-to-day lives.

When you immerse yourself in the environment and in the *context* of the people you design for, you can often get a completely different perspective than what you might expect. What might seem like good weather for you (say, warm and sunny) may well be a not-so-nice day for someone else.

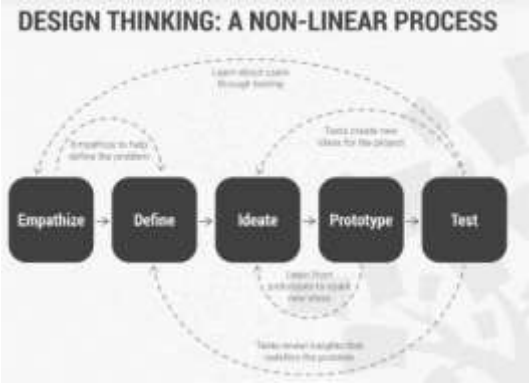
To ‘Empathize’ is simply crucial to a human-centered design process such as design thinking as it helps you set aside your own assumptions about the world to gain insight into your users and their needs instead. The Empathize stage of the design thinking process is a time for you to collect as many experiences, insights and observations as possible, so you can build a solid foundation for the rest of your design project. Experiences, insights and observations as possible, so you can build a solid foundation for the rest of your design project.

We cannot stress enough how important it is for you as a designer to develop the best possible understanding of your users, their needs and the problems that your product or service is trying to solve! You and your team can gain huge insight via the practical Empathize methods below, and, if time and money allows for it, it may also be worth getting some experts on board to help you fully understand the people you want to design for.



What’s good to note is this empathic phase of design thinking is named differently depending on which version of the methodology you follow. Different schools and companies which use design thinking have called empathic research "the Empathize stage" (as we do), "the understand phase," "the hear phase," and simply "looking," as well as a number of other terms. Regardless of which phrase you’re familiar with, the core is essentially the same — **empathy is deeply human-centric and is essential at the start of any design process.**

Empathize is design thinking's first stage for a reason. It's the first step on the road to thoughtfully designed products that prove the designers built with a compassionate eye for their users. Empathy is a naturally occurring characteristic which people have in varying degrees. However, they can improve their ability to empathize as a soft skill. Anyone in a design team will have preconceived ideas about the many situations people find themselves in as users. It's unavoidable – you can't unlearn your life experience. Therefore, you should always **adopt a beginner's mindset to be able to view and analyze situations with users objectively**.



To empathize is to research. So, you should constantly remind yourself to question everything you observe instead of judging. You should also listen to others open-mindedly rather than focus on points that confirm your biases. Because our biases will naturally creep into how we view the world and the situations we consider, as designers—or design thinkers—we must catch and overcome these before they distort our research. You must become fully objective before you can start to see through your users' eyes and interpret their viewpoints optimally. They are the experts. You must understand the users' dimensions of use (e.g., tasks) and their feelings (e.g., motivations) before you can work towards delighting them through your design.

There are different way to empathize to get the task going on right track:

1. **Observing real users:** Ask these questions to shift from concrete observations to abstract motivations:
    a. "What?" – You detail your observations.
    b. "How?" – You analyze how users do things (e.g., with difficulty).
    c. "Why?" – You make educated guesses about the users' emotions and motivations.
2. **Conducting photo**- and/or video-based studies in users' natural environments or sessions with the design team or consultants – You record these users while they try to solve an issue you propose to resolve with your design.
3. **Personal photo/video journals** – You ask users to record their own experiences with approaching a problem. These may capture their pain points more accurately.
4. **Interviewing users** – Your team uses brainstorming to first find the right questions to ask in a generally structured and natural flow. Then, you can directly ask users for their insights in an intimate setting where they can respond earnestly to open-ended questions.
5. **Engaging with extreme users** – You find the extreme cases within your userbase to determine the greatest degrees of users' needs, problems and problem-solving methods. You can then see the full scope of problems which typical, non-extreme users might run into. If you can satisfy an extreme user, you can satisfy any user.

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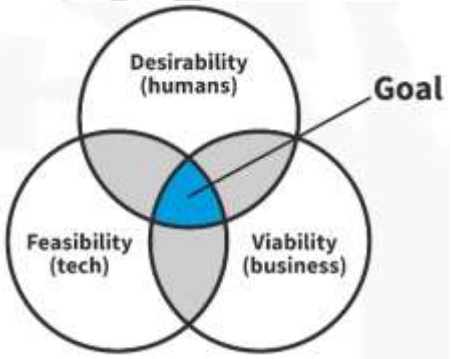
6. **Analogous empathy** – Your team finds effective analogies to draw parallels between users' problems and problems in other fields. This way, you can get insights you'd otherwise overlook.
7. **Sharing inspiring stories** – Your team shares stories about what they have observed so you draw meaning from these and note fascinating details.
8. **Bodystorming** – You wear equipment (e.g., goggles, gloves, torso attachments) to gain first-hand experience of your users in their environment.
9. **Empathy maps and customer journey maps** – Your team should have at least one of these as a reference point to appreciate the users' perspectives.
10. Personas to establish accurate portraits/profiles of users who'll interact with your product.

Q. 3 Why Empathy is crucial for Business Success?

Empathy can also be deemed an essential component of business solutions when you look at things from the perspective of profit. You may create solutions which completely miss the mark if you develop solutions in isolation — you need to gain essential insights about your users if you want to remain relevant in the market.

Many leaders within the fields of innovation, learning and entrepreneurship have pointed to three key parameters which define a successful product or service: desirability, feasibility and viability.

It's not enough that the technology exists (i.e., feasibility is present) and that profits or business benefits may be derived (i.e., it is viable). Users need to feel a sense of desirability towards a solution. We can only design a desirable product or service when people's needs, experiences, wants and preferences are properly understood.



Example of Empathy in Business:

One clear example of this is the iPod. Many MP3 players came and went throughout the late 1990s, and didn't create much of an impact. Then, along came the iPod in 2001. It not only provided a technological solution, but also delivered a completely desirable and viable experience. This meant Apple took the lead in the market and continued to hold it for many years to come, generating huge profits along the way.

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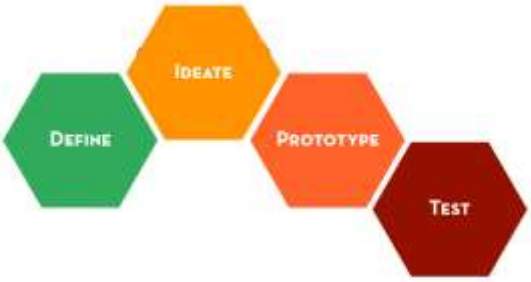
Q. 4. Explain all the stages of Empathy:



Stage 1: Define

WHAT is the Define mode?

The Define mode of the design process is all about bringing clarity and focus to the design space. It is your chance, and responsibility, as a design thinker to define the challenge you are taking on, based on what you have learned about your user and about the context. After becoming an instant-expert on the subject and gaining invaluable empathy for the person you are designing for, this stage is about making sense of the widespread information you have gathered. The goal of the Define mode is to craft a meaningful and actionable problem statement – this is what we call a point-of-view. This should be a guiding statement that focuses on insights and needs of a particular user, or composite character. Insights don't often just jump in your lap; rather they emerge from a process of synthesizing information to discover connections and patterns. In a word, the Define mode is sense making

WHY Define?

The Define mode is critical to the design process because it results in your point-of-view (POV): the explicit expression of the problem you are striving to address. More importantly, your POV defines the RIGHT challenge to address, based on your new understanding of people and the problem space. It may seem counterintuitive but crafting a more narrowly focused problem statement tends to yield both greater quantity and higher quality solutions when you are generating ideas. The Define mode is also an endeavor to synthesize your scattered findings into powerful insights. It is this synthesis of your empathy work that gives you the advantage that no one else has: discoveries that you can leverage to tackle the design challenge; that is, INSIGHT.

HOW to Define?

Consider what stood out to you when talking and observing people. What patterns emerge when you look at the set? If you noticed something interesting ask yourself (and your team) why that might be. In asking why someone had a certain behavior or feeling you are making connections from that person to the larger context. Develop an understanding of the type of person you are designing for – your USER. Synthesize and select a limited set of NEEDS that you think are important to fulfill; you may in fact express a just one single salient need to address. Work to express INSIGHTS you developed through the synthesis of information you have gathered through empathy and research work. Then articulate a point-of-view by combining these three elements – user, need, and insight – as an actionable problem statement that will drive the rest of your design work. A good point-of-view is one that: - Provides focus and frames the problem - Inspires your team - Informs criteria for evaluating competing ideas - Empowers your team to

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make decisions independently in parallel - Captures the hearts and minds of people you meet - Saves you from the impossible task of developing concepts that are all things to all people (i.e. your problem statement should be discrete, not broad.)

Stage 2: Ideate

WHAT is the Ideate mode?

Ideate is the mode of the design process in which you concentrate on idea generation. Mentally it represents a process of "going wide" in terms of concepts and outcomes. Ideation provides both the fuel and also the source material for building prototypes and getting innovative solutions into the hands of your users.

WHY ideate?

You ideate in order to transition from identifying problems to creating solutions for your users. Ideation is your chance to combine the understanding you have of the problem space and people you are designing for with your imagination to generate solution concepts. Particularly early in a design project, ideation is about pushing for a widest possible range of ideas from which you can select, not simply finding a single, best solution. The determination of the best solution will be discovered later, through user testing and feedback. Various forms of ideation are leveraged to: - Step beyond obvious solutions and thus increase the innovation potential of your solution set - Harness the collective perspectives and strengths of your teams - Uncover unexpected areas of exploration - Create fluency (volume) and flexibility (variety) in your innovation options - Get obvious solutions out of your heads, and drive your team beyond them

HOW to ideate?

You ideate by combining your conscious and unconscious mind, and rational thoughts with imagination. For example, in a brainstorm you leverage the synergy of the group to reach new ideas by building on others' ideas. Adding constraints, surrounding yourself with inspiring related materials, and embracing misunderstanding all allow you to reach further than you could by simply thinking about a problem. Another ideation technique is building – that is, prototyping itself can be an ideation technique. In physically making something you come to points where decisions need to be made; this encourages new ideas to come forward. There are other ideation techniques such as bodystorming, mindmapping, and sketching. But one theme throughout all of them is deferring judgment – that is, separating the generation of ideas from the evaluation of ideas. In doing so, you give your imagination and creativity a voice, while placating your rational side in knowing that your will get to the examination of merits later.

Stage 3: Prototype

WHAT is the Prototype mode?

The Prototype mode is the iterative generation of artifacts intended to answer questions that get you closer to your final solution. In the early stages of a project that question may be broad – such as "do my users enjoy cooking in a competitive manner?" In these early stages, you should create low-resolution prototypes that are quick and cheap to make (think minutes and cents) but can elicit useful feedback from users and colleagues. In later stages both your prototype and question may get a little more refined. For example, you may create a later stage prototype for the cooking project that aims to find out: "do my users enjoy cooking with voice commands or visual commands". A prototype can be anything that a user

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can interact with – be it a wall of post-it notes, a gadget you put together, a role-playing activity, or even a storyboard. Ideally you bias toward something a user can experience. Walking someone through a scenario with a storyboard is good, but having them role-play through a physical environment that you have created will likely bring out more emotions and responses from that person.

WHY prototype?

To ideate and problem-solve. Build to think. To communicate. If a picture is worth a thousand words, a prototype is worth a thousand pictures. To start a conversation. Your interactions with users are often richer when centered on a conversation piece. A prototype is an opportunity to have another, directed conversation with a user. To fail quickly and cheaply. Committing as few resources as possible to each idea means less time and money invested up front. To test possibilities. Staying low-res allows you to pursue many different ideas without committing to a direction too early on. To manage the solution-building process. Identifying a variable also encourages you to break a large problem down into smaller, testable chunks.

HOW to prototype?

Start building. Even if you aren't sure what you're doing, the act of picking up some materials (post-its, tape, and found objects are a good way to start!) will be enough to get you going. Don't spend too long on one prototype. Let go before you find yourself getting too emotionally attached to any one prototype. ID a variable. Identify what's being tested with each prototype. A prototype should answer a particular question when tested. That said, don't be blind to the other tangential understanding you can gain as someone responds to a prototype. Build with the user in mind. What do you hope to test with the user? What sorts of behavior do you expect? Answering these questions will help focus your prototyping and help you receive meaningful feedback in the testing phase.

Stage 4: Test

WHAT is the Test mode?

The Test mode is when you solicit feedback, about the prototypes you have created, from your users and have another opportunity to gain empathy for the people you are designing for. Testing is another opportunity to understand your user, but unlike your initial empathy mode, you have now likely done more framing of the problem and created prototypes to test. Both these things tend to focus the interaction with users, but don't reduce your "testing" work to asking whether or not people like your solution. Instead, continue to ask "Why?", and focus on what you can learn about the person and the problem as well as your potential solutions. Ideally you can test within a real context of the user's life. For a physical object, ask people to take it with them and use it within their normal routines. For an experience, try to create a scenario in a location that would capture the real situation. If testing a prototype in situ is not possible, frame a more realistic situation by having users take on a role or task when approaching your prototype. A rule of thumb: always prototype as if you know you're right, but test as if you know you're wrong—testing is the chance to refine your solutions and make them better.

WHY test?

To refine prototypes and solutions. Testing informs the next iterations of prototypes. Sometimes this means going back to the drawing board. To learn more about your user. Testing is another opportunity to build empathy through observation and engagement—it often yields unexpected insights. To refine your

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POV. Sometimes testing reveals that not only did you not get the solution right, but also that you failed to frame the problem correctly.

HOW to test?

Show don't tell. Put your prototype in the user's hands – or your user within an experience. And don't explain everything (yet). Let your tester interpret the prototype. Watch how they use (and misuse!) what you have given them, and how they handle and interact with it; then listen to what they say about it, and the questions they have. Create Experiences. Create your prototypes and test them in a way that feels like an experience that your user is reacting to, rather than an explanation that your user is evaluating. Ask users to compare. Bringing multiple prototypes to the field to test gives users a basis for comparison, and comparisons often reveal latent needs.

Q. 5. Empathy Canvas. Explain Empathy canvas for a restaurant management system.

Empathy Canvas is divided into various sections the section which you are seeing over your is a user section then the stakeholder's, activities, storyboarding, and then you have a happy sad story.

1. User:

Users are those people, who are directly or indirectly attached to your system, who will be using your system. In a typical restaurant management system, users include: customers, managers, billdesk employees, chefs, waiters, inventory personal etc.

2. Stakeholders:

Stakeholders are those, who benefit from your system, which include: restaurant owners, waiters (getting tip), chefs (getting feedback), inventory personal (getting account of the stock), and managers (feedback regarding service improvement).

Stakeholders are those who are going to get a direct or an indirect income because of these users so write down the number n number of names of these stakeholders in this particular column of stakeholders

3. Activities:

Activities are those tasks which users and stakeholders will be doing by using your system. In a typical restaurant management system activities include:

- Customer: browses the menu, orders the food, makes payment, gives feedback, reserves table etc.
- Chef: receives order, intimates waiter, checks inventory, analyzes food feedback etc.
- Bill Desk Manager: receives payment, manages salary, monthly/weekly cashflow etc

4. Storyboard:

Story board is that section where, designer will try to create an imaginary story on how day to day your system will be used. In case of restaurant management system, a daily activity of a restaurant can be portrayed as a story. A worst case scenario can also be portrayed as a section of sad story.

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Q. 6. Unified Modeling Language (UML) diagrams – Behavioral Diagrams

UML stands for **U**nified **M**odeling **L**anguage. It's a rich language to model software solutions, application structures, system behavior and business processes.

Behavioral Diagrams:

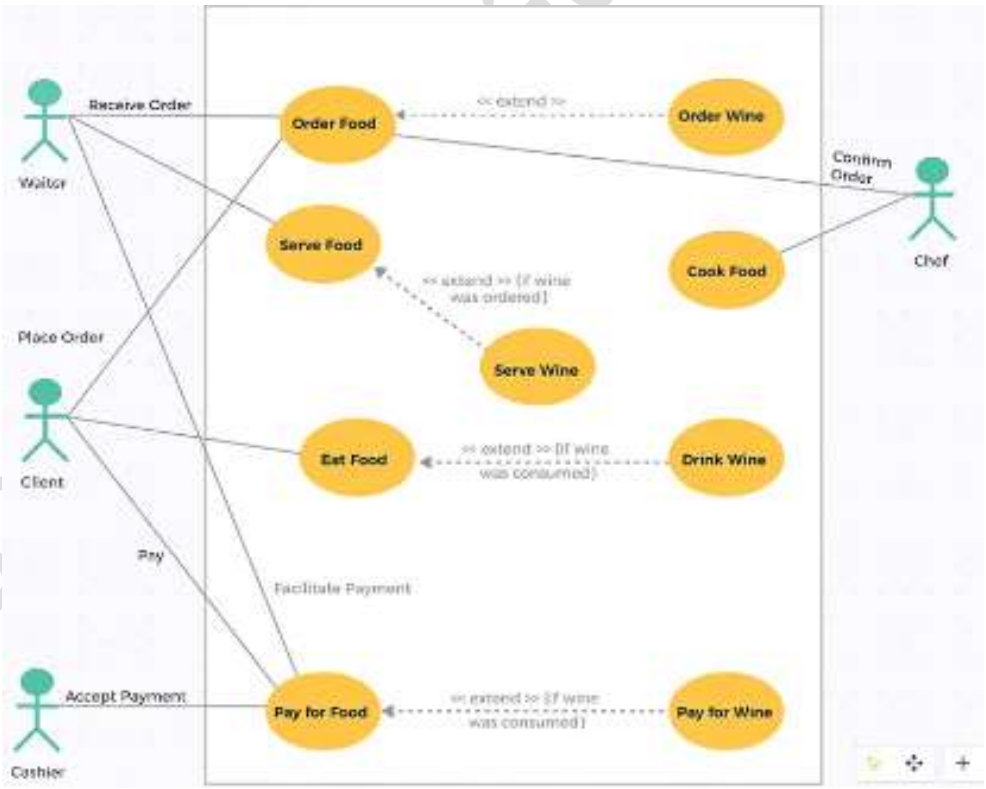
- Use Case Diagram
- Activity Diagram
- State Machine Diagram

Use Case Diagram:

As the most known diagram type of the behavioral UML types, Use case diagrams give a graphic overview of the actors involved in a system, different functions needed by those actors and how these different functions interact.

A use case diagram specifies;

- The system
- The actors (those who interact with the system)
- The use cases (what the actors can do)



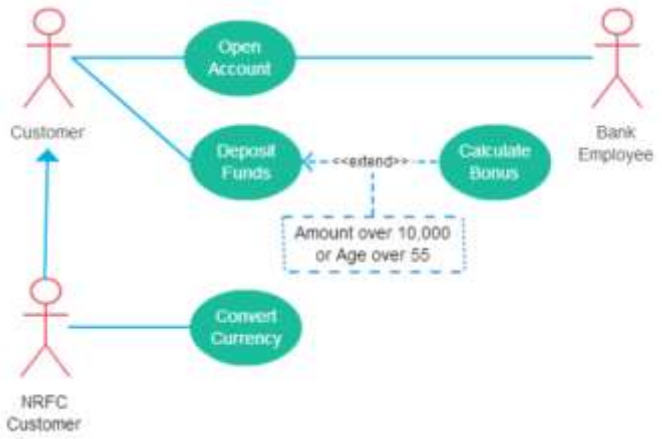
Types of relations in use case:

There are two types of relations in uses cases: include and extend

1. Extend Relation

As the name implies it extends the base use case and adds more functionality to the system. Here are a few things to consider when using the <<extend>> relationship.

- The extending use case is dependent on the extended (base) use case. In the below diagram the "Calculate Bonus" use case doesn't make much sense without the "Deposit Funds" use case.
- The extending use case is usually optional and can be triggered conditionally. In the diagram, you can see that the extending use case is triggered only for deposits over 10,000 or when the age is over 55.
- The extended (base) use case must be meaningful on its own. This means it should be independent and must not rely on the behavior of the extending use case.



Although extending use case is optional most of the time it is not a must. An extending use case can have non-optional behavior as well. This mostly happens when you're modeling complex behaviors.

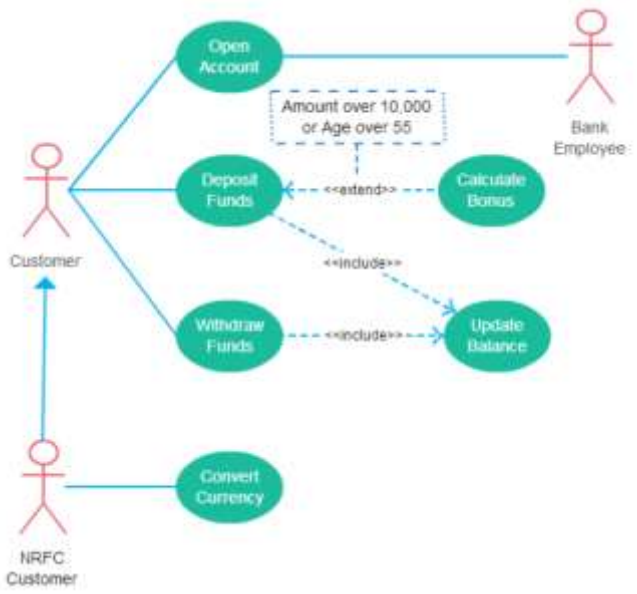
For example, in an accounting system, one use case might be "Add Account Ledger Entry". This might have extending use cases "Add Tax Ledger Entry" and "Add Payment Ledger Entry". These are not optional but depend on the account ledger entry. Also, they have their own specific behavior to be modeled as a separate use case.

2. Include Relation:

Include relationship show that the behavior of the included use case is part of the including (base) use case. The main reason for this is to reuse common actions across multiple use cases. In some situations, this is done to simplify complex behaviors. Few things to consider when using the <<include>> relationship.

- The base use case is incomplete without the included use case.
- The included use case is mandatory and not optional.

Let expand our banking system use case diagram to show include relationships as well.



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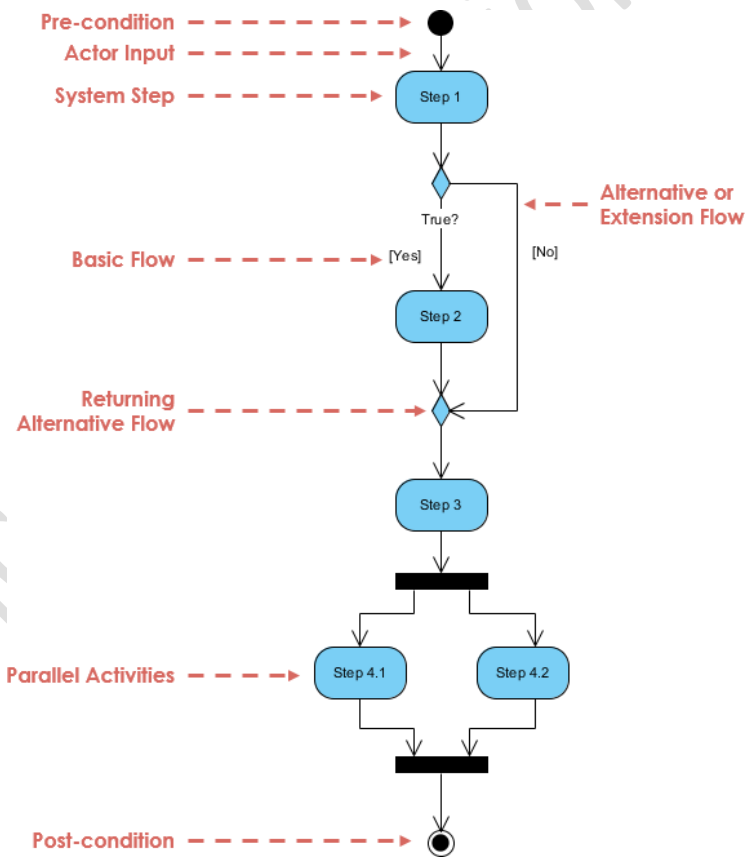
Activity Diagrams:

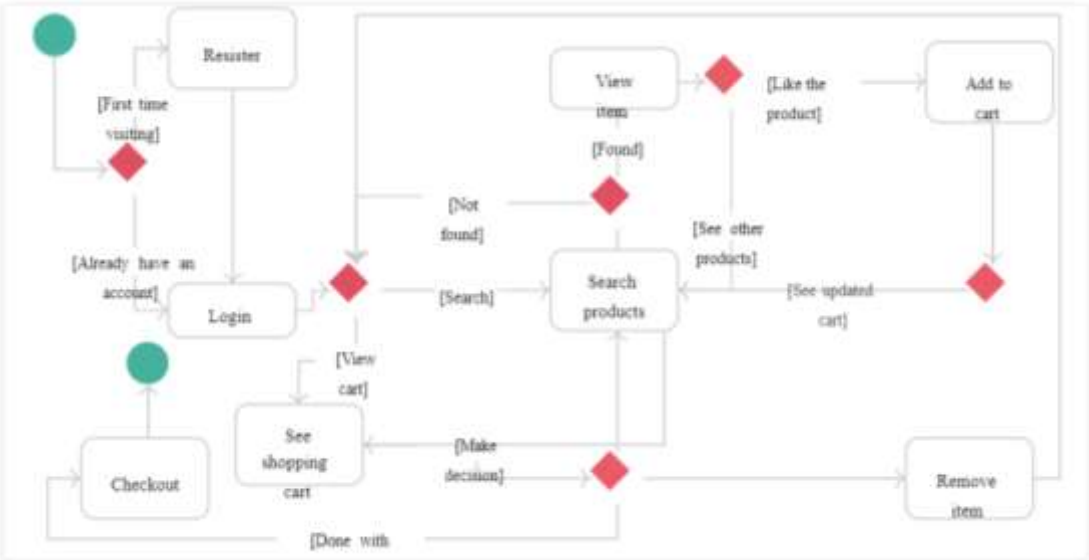
Activity diagrams represent workflows in a graphical way. They can be used to describe the business workflow or the operational workflow of any component in a system.

Activity Diagrams describe how activities are coordinated to provide a service which can be at different levels of abstraction. Typically, an event needs to be achieved by some operations, particularly where the operation is intended to achieve a number of different things that require coordination, or how the events in a single use case relate to one another, in particular, use cases where activities may overlap and require coordination. It is also suitable for modeling how a collection of use cases coordinate to represent business workflows

- Identify candidate use cases, through the examination of business workflows
- Identify pre- and post-conditions (the context) for use cases
- Model workflows between/within use cases
- Model complex workflows in operations on objects
- Model in detail complex activities in a high level activity Diagram



State Machine Diagram

State machine diagrams are similar to activity diagrams, although notations and usage change a bit. They are sometimes known as state diagrams or state chart diagrams as well. These are very useful to describe the behavior of objects that act differently according to the state they are in at the moment. The State machine diagram below shows the basic states and actions.

State Diagram for Online Shopping System